Changes in Arctic Wintertime Surface-based Inversions

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1 Methods

1.1 Datasets

Reanalysis and model monthly average data were used to analyze surface-based inversion climatology and trends. The reanalysis dataset chosen was ERA5, which is the fifth generation ECMWF atmospheric reanalysis of the global climate [4]. The models were were from the Coupled Model Intercomparison Project phase 6 (CMIP6). The models selected were the Canadian Earth System Model version 5 (CanESM5) from Canadian Centre for Climate Modelling and Analysis (CCCma)[6], and IPSL-CM6A-LR from the Insitut Pierre Simon Laplace [2]. The model experiments used were AMIP, which is a historical Atmospheric Model Intercomparison Project simulation where sea surface temperatures and sea ice concentrations are prescribed based on observations[5]; and historical, which is a simulation of the historical period forced, based on observations, by evolving externally imposed forcings such as solar variability, volcanic aerosols, and changes in atmospheric composition caused by human activities [5]. These five datasets were compared to see how well the model and reanalysis data agree with each other. Pressure level and surface data were obtained for years 1979-2014, which are the years available for the AMIP experiment; for the ERA5 data, this was also extended to 2024. Only the winter months, December, January and February, were analyzed. Calculations were done for the entire northern hemisphere, but here the results focus on the Arctic, here defined as 66.5°N to 90°N.

1.2 Inversion Quantification

Surface-based inversions exist when temperature increases with height from the surface. Two metrics were used to quantify surface-based inversions: intensity and depth, defined as the temperature difference between the top of the inversion layer and the surface [7]:

$$\Delta T = T(z_{top}) - T(z_{sfc}) \tag{1}$$

and the depth, Δz. To convert the pressure levels to depth in meters, the hydrostatic equation was used:

$$\frac{\partial p}{\partial z} = -\rho g \tag{2}$$

where p is pressure, z is altitude, ρ is the density of air, and g is the gravitational acceleration. Integrating and substituting the ideal gas law $p = \rho R' T$ written in terms of the gas constant for dry air and virtual temperature $p = \rho_m R_d T_v$ gives the hypsometric equation, where $\overline{T_v}$ is the layer mean virtual temperature, w is the water vapour mixing ratio, $R_d = 287 \frac{J}{kgK}$ is the gas constant for dry air, and $g_o = 9.8\ m/s^2$ is the gravitational acceleration constant:

$$\Delta z = \frac{R_d \overline{T_v}}{g_o} ln \frac{p_1}{p_2} \approx \frac{R_d(1 + 0.6w)\overline{T}}{g_o} ln \frac{p_1}{p_2} \tag{3}$$

In this analysis, the specific humidity is assumed to be approximately equal to, and used in place of, the water vapour mixing ratio. Going layer by layer from the surface, the depth of each layer was calculated by using the average values of temperature and specific humidity between two pressure levels. Each layer was added to the preceding layers up until the inversion top level to obtain the total inversion depth. Points with no inversion were given a value of 0 for intensity an depth.

Using the discrete pressure level data, with a resolution of 25 hPa in the lower troposphere, the surface temperature was compared to the temperature at the first pressure level that is less (of greater altitude) than the surface pressure. The inversion top was defined as the highest (in altitude) level whose temperature is greater than or equal to the level below[3]. Isothermal layers at the base, top, or within an inversion layer were included [3]. This definition assumes there are no inflection points within a deeper inversion layer; often layers of decreasing temperature with height are considered to still be part of an inversion layer if they are less than 100 m deep [1][7]. However, the pressure levels are more than 100 m apart: taking the greatest values for the ERA5 pressure levels, 1000 hPa and 975 hPa, which minimizes the logarithm term in Equation 3, the temperature would still have to be around 134K or less for the depth to be less than 100m, even assuming completely dry air. Since this temperature is not a physically reasonable value in the atmosphere, small layers of positive lapse rate do not change the analysis using these data. Depending on the surface pressure, the only level that could be less than 100 m deep is the one right above the surface. However, this means that the height of the inversion top is not precise as it could be between pressure levels which are far apart. This has a bigger effect on the model data; the models have 19 pressure level coordinates (5 levels below 500 hPa), while ERA5 has 37 (15 levels below 500 hPa). The models may overestimate inversion intensity and depth compared to the reanalysis but may not be able to detect shallow inversions.

1.3 Trends

To test whether there is a significant trend, either increasing or decreasing, at $\alpha = 0.05$, a linear least-squares regression was performed on the intensity and depth timeseries from 1979-2014 at each grid point using the Python library *scipy.stats.linregress*.

To examine how changes in the vertical temperature profile may be causing these trends, further analysis was done at selected points. The points used were the points of greatest increasing and decreasing trends for intensity and depth across all the winter months according to the ERA5 data. In addition to these single grid points, area averages were also examined for two regions of interest north of Greenland (83°N to 90°N, 0°to 100°W) in February, and around the Chukchi sea (69°N to 75°N, 150°to 180°W) in December. The area averages were done by first taking the zonal mean, and then taking the cosine-weighted average across latitudes.

At each point, time-series from 1979-2014 for that month were plotted, and a linear least-squares regression was performed for the inversion top temperature and surface temperature. Additionally, for each point, a trend profile was plotted that shows the slope of the linear regression on the time series for each pressure level. In this case, the slope was plotted whether or not the regression is significantly different from zero. The depth time and inversion top pressure level time series were also plotted for reference when looking at vertical profile changes. To compare with the models, which have lower grid resolutions, the point closest to the corresponding point in the ERA5 grid was used.

2 Results

2.1 Climatology

The climatologies for each month and each dataset were calculated by taking the time mean at each grid point.

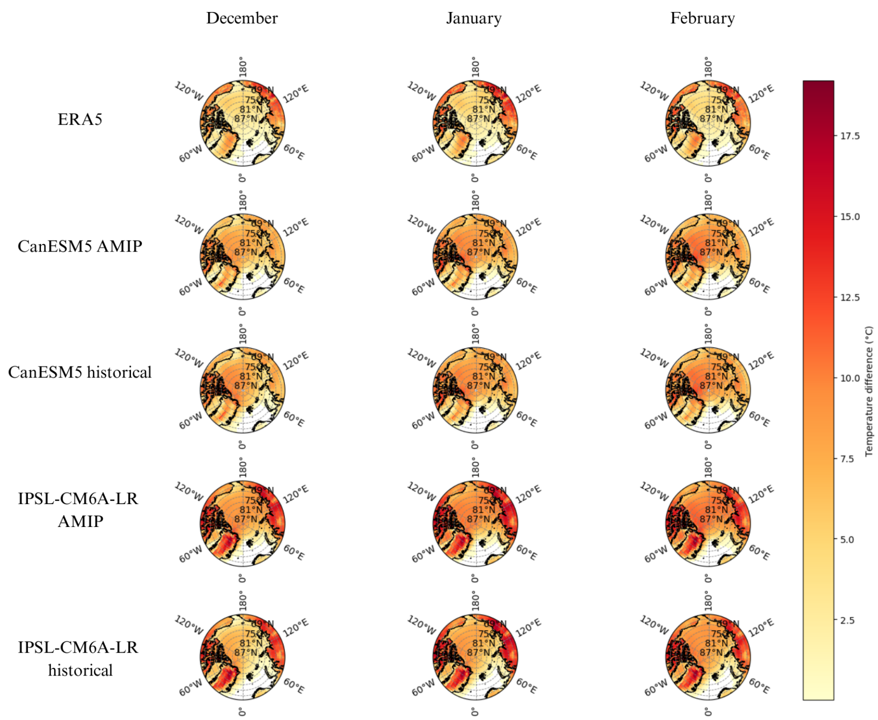

Figure 1: 1979-2014 mean T(inversion top) - T(surface)

All the datasets agree that there is on average no inversion in the Barents Sea region. The spatial distribution are similar across all datasets and are generally consistent with previous research by others, for example Zhang et al. (2011) [7]. Some differences between datasets are that the models have stronger intensity over the Arctic Ocean than ERA5, and both IPSL-CM6A-LR experiments show strong intensity in the land regions, while CanESM5 is more homogeneous across land and ocean. The climatologies are also similar across all the winter months.

Note that since zero values were assigned to grid/time points with no inversion, points showing stronger intensity on average may not necessarily mean the intensity themselves are greater, but that they are more frequent. Having not taken into account inversion frequency in this analysis, one simple way to estimate the relative frequencies is by comparing to a plot which uses NaN values instead of zeros for points with no inversion. In that case, a NaN mean was taken, where the NaN points were skipped over and the mean is only over time points where there is inversion. The larger the resulting intensity is compared to the previous intensities using zero values, the less frequent the inversions are. This plot using NaN values are shown in Appendix A. . The resulting plots are very similar to figure 1, with some slightly stronger intensities in certain regions, notably the Arctic Ocean in ERA5 and the Barents Sea for CanESM5 amip January and February, and both IPSL-CM6A-LR experiments for all months. This shows that in these areas there are some time points with no inversion, but overall, where there is inversion at all in figure 1, it exists at almost every time point, which is not surprising since the data used is the monthly average.

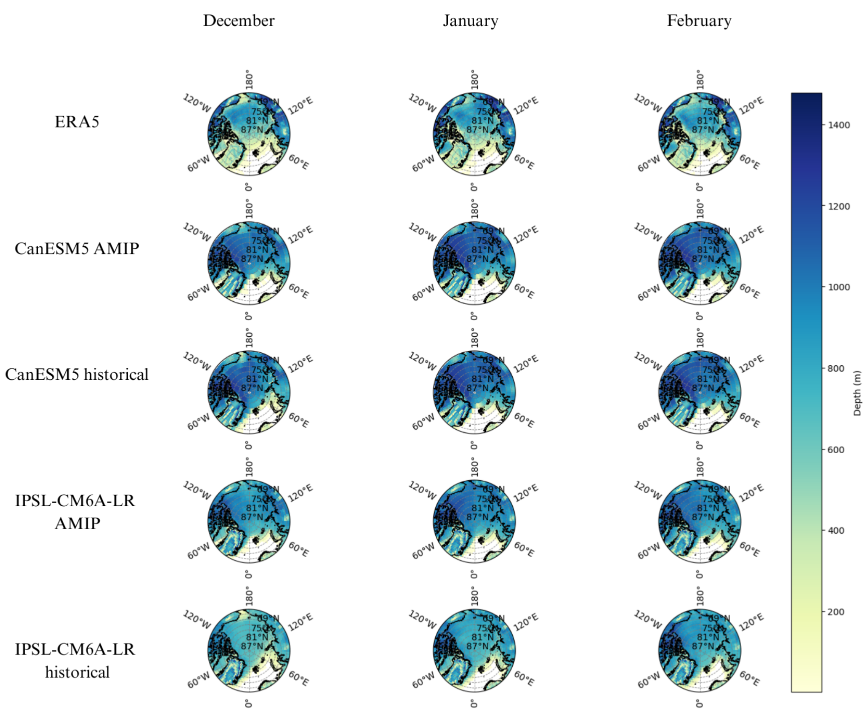

Figure 2: 1979-2014 mean inversion depth

The inversion depths are in general more homogeneous across land and ocean. There appears to be unusual discontinuities over Greenland for all the datasets, and further investigation is needed to determine the cause.

Extending to 2024 for the ERA5 data, the climatology maps for inversion and depth are visually identical to those up to 2014 (see Appendix B. ; a side-by-side comparison using NaN values is also included).

2.2 Trends

The trend maps show the slope of the linear regression on the time series at each grid point. Trends that are not significant at $\alpha = 0.05$ are masked out.

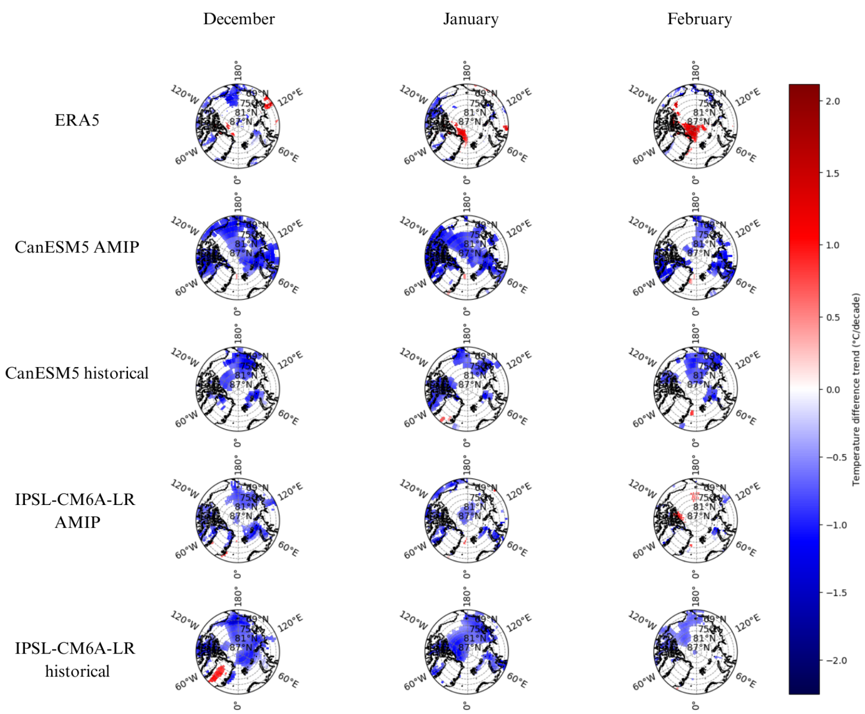

Figure 3: 1979-2014 inversion intensity trends

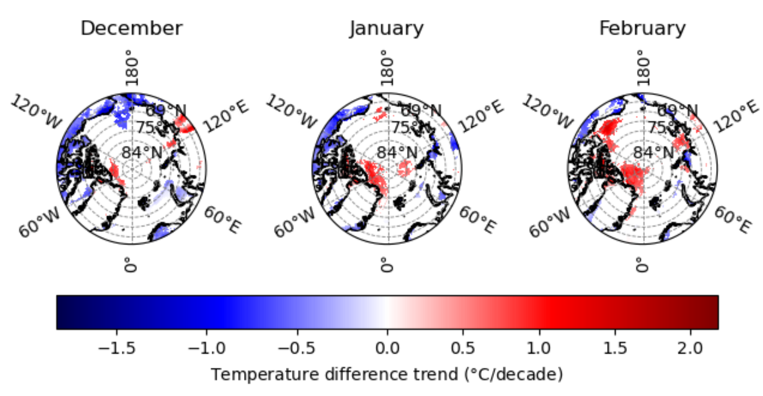

Figure 4: 1979-2024 ERA5 inversion intensity trends

There are large variations in inversion intensity trends between months and datasets. The models generally show much larger areas of decreasing inversion intensity over the Arctic Ocean. Trends that are generally consistent across datasets include the decreasing trends in the area around the Chukchi Sea and northward in December, Nunavut in December and January, and north of Novaya Zemlya in December. The areas of increasing trend are inconsistent across datasets; notable areas are north of Greenland in January and February in the ERA5 dataset, and Greenland in the IPSL-CM6A-LR dataset.

Extending to 2024 for the ERA5 data, the trends are similar to those up to 2014, with some expansion of areas of decreasing trend over land areas and areas of increasing trend over ocean.

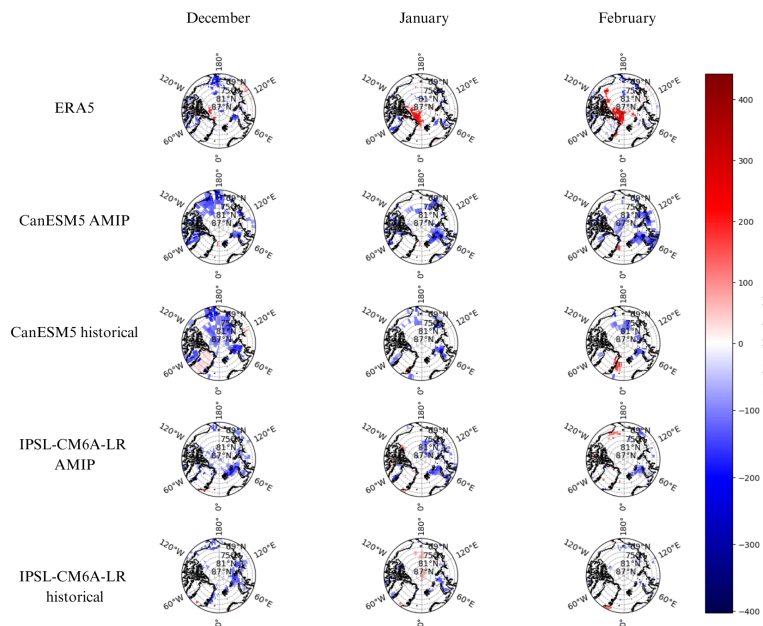

Figure 5: 1979-2014 inversion depth trends

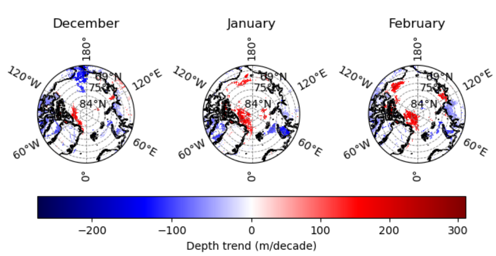

Figure 6: 1979-2024 ERA5 inversion depth trends

There are in general less significant trends in inversion depth than intensity. Areas of trends that somewhat agree across datasets also correspond to the previously mentioned areas of intensity trend; northward of the Chukchi Sea and north of Novaya Zemlya in December. The large areas of increasing trend north of Greenland in January and February in the ERA5 dataset also corresponds to the similar area in the intensity trend maps.

Extending to 2024 for the ERA5 data, the trends are again similar to those up to 2014, with the most notable changes being increases in areas of increasing trend over ocean areas, particularly in February.

2.3 Time series and trends at select points

Each of the points of greatest trend according to the ERA5 data, both increasing and decreasing, occur in February. The point of greatest increasing intensity at a rate of 2.1K temperature difference/decade is 83.25°N, 49°W, off the north coast of Greenland, and the point of greatest increasing depth at a rate of 440 m/decade is 75.75°N, 110.5°W, off the coast of Melville Island in Nunavut. The point of greatest decreasing intensity at a rate of -2.3K/decade is 67.25°N, 176.25°W, inland of Vankarem on the eastern tip of Russia, and the point of greatest decreasing depth at a rate of -400m/decade is 69°N, 179.5°W, off the coast of Ryrkaypiy on the eastern tip of Russia, relatively close to the point of greatest decreasing intensity. Note these are the specific coordinates; the ERA5 data has a grid resolution of 0.25°latitude × 0.25°longitude, while the CanESM5 data have a grid resolution of around 2.8°latitude × 2.8°longitude, and the IPSL-CM6A-LR data 1.3°latitude × 2.5°longitude, so the model grid points cover more area. For the time series, points with no inversion are given inversion top values equal to the surface values.

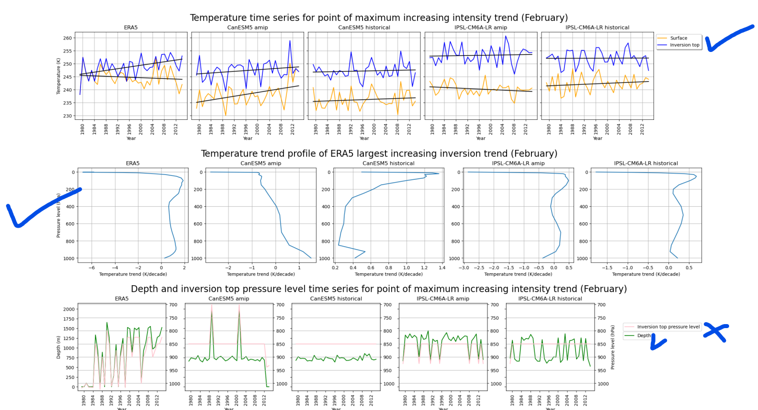

Figure 7: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series for the point of greatest increasing intensity trend (83.25 °N, 49 °W in February)

From the ERA5 trend profile in the second plot, it is seen that the trend corresponding to the inversion top pressure level in the bottom plot (between 500 hPa-800 hPa where it exists) has a greater positive value than near the surface, hence the increasing intensity trend. From the inversion top and surface time series, it is seen that there is no inversion in the first few years, then the inversion top temperature increases significantly with p=0.003 and $r^2 = 0.2$ at a rate of 1.7K/decade, while there is no significant trend for the surface temperature.

The datasets are not in agreement at this point. The only other temperature time series with a significant trend is the CanESM5 amip surface temperature, with p=0.006 and $r^2 = 0.2$ with a slope of 1.8K/decade; therefore that dataset actually shows a decreasing intensity trend. The ERA5 data also has generally higher surface temperatures than the models at this point.

The depth time series show that it is generally not accurate to use a linear regression on inversion depth, since there are often spikes that would not be so large if more continuous measurements were used rather than the discrete pressure levels.

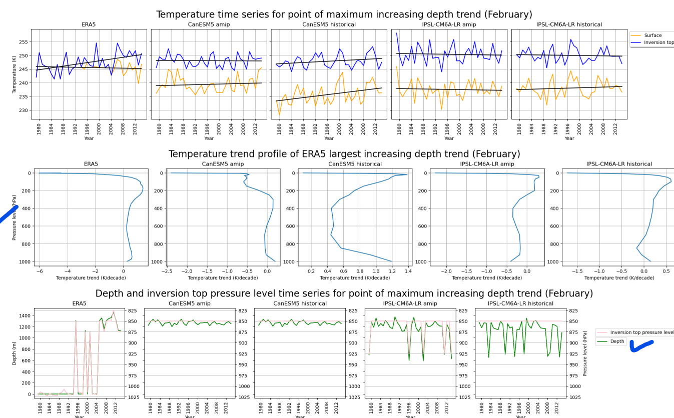

Figure 8: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series for the point of greatest increasing depth trend (75.75°N, 110.5°W in February)

From the temperature and depth time series, it is seen that there are on average no inversions in February at this point over the first 10 years, with sporadic inversions through the middle period and consistent inversions in the last few years. Again, the datasets do not agree at this point; the models all show an inversion at all time points, and again the surface temperature is in general higher in the ERA5 data than the models. The only timeseries that show a significant temperature trend are the ERA5 inversion top, with p=0.002, $r^2 = 0.2$, and a slope of 1.6K/decade, and the CanESM5 historical surface, with p=0.01, $r^2 = 0.2$, and a slope of 1.4K/decade, which result in opposing inversion trends.

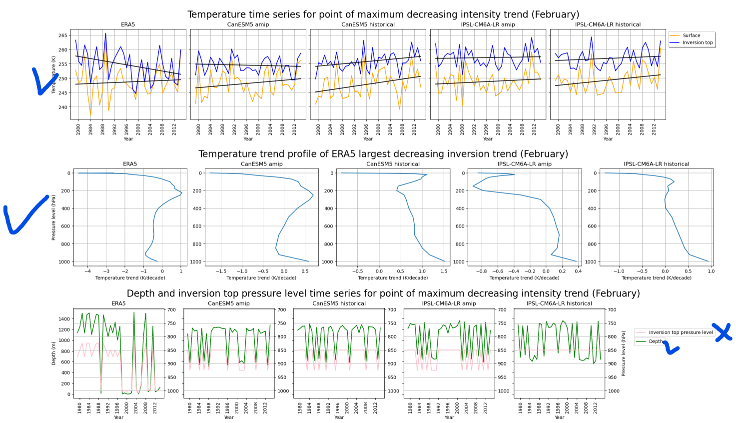

Figure 9: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series for the point of greatest decreasing intensity trend (67.25°N, 176.25°W in February)

From the ERA5 trend profile, it is seen that the trend corresponding to the inversion top pressure level in the bottom plot (between 825 hPa-875 hPa where it exists) has a greater negative value than near the surface (note that the surface data is not actually accounted for in the trend profile if the surface pressure is greater than 1000 hPa), hence the decreasing intensity trend. The datasets are more in agreement in terms of the shape of the trend profile, with near surface trends exceeding inversion top trends; however, it is important to note that the ERA5 dataset actually shows a decrease in temperature while the models show an increase in temperature near the surface, with CanESM5 amip temperatures beginning to decrease around 900 hPa.

From the inversion top and surface temperature time series, the inversion top temperature decreases significantly with p=0.003 and $r^2 = 0.1$ at a rate of -1.8K/decade, while there is no significant trend for the surface temperature. There are no significant trends for the other temperatures, except the CanESM5 historical surface temperature, with p=0.04, $r^2 = 0.1$, and a slope of 1.6K/decade. Therefore both these datasets show decreasing intensity, but the ERA5 is due to decreasing inversion top temperature while the CanESM5 historical is due to increasing surface temperature.

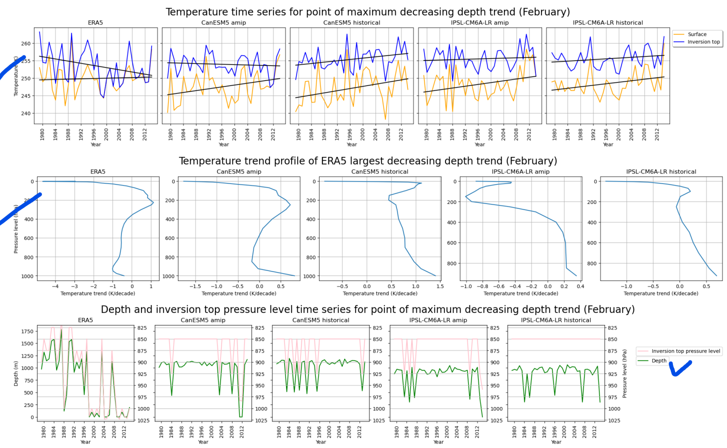

Figure 10: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series for the point of greatest decreasing depth trend (69°N, 179.5°W in February)

The trend profiles are similar to those of the decreasing intensity trend from figure 9. Again, the ERA5 dataset gives decreasing temperature trends while the models show increasing trends. In terms of inversion depth, there is a large drop-off around 1998, with having no/shallow inversions with two large spikes. The ERA5 inversion top and surface temperatures themselves actually do not have a significant trend; the only temperature time series that has a significant trend is the CanESM5 historical surface tmperature, with p=0.04, $r^2 = 0.1$, and a slope of 1.5K/decade. The models do not show any significant trends in inversion depth.

Next, two area averages were selected for large areas of increasing and decreasing inversion intensity respectively in the ERA5 data: 83°N to 90°N, 0°to 100 °W, to the north of Greenland, in February, and 69°N to 75°N, 150°W to 180 °around the Chukchi Sea, in December. Note that the inversion top pressure levels were determined from the area weighted average, while the depth values were first calculated for each point then averaged.

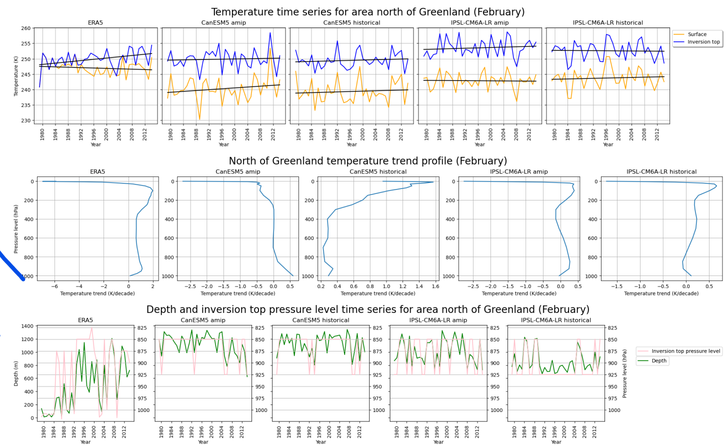

Figure 11: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series north of Greenland in February)

The results are similar to those of the single point from figure 7, which is also part of this area average. The ERA5 trend profile is similar to that of the IPSL-CM6A-LR amip profile; however, in general the datasets do not agree, and the surface temperatures for ERA5 are again higher than the models. The inversion top temperature time series for the ERA5 data has a significant increasing trend with p = 0.02, $r^2 = 0.1$, and a slope of 1.0K/decade. None of the other temperature time series show a significant trend.

The depth time series appear to be smoothed out compared to the ones at single points, with less sharp spikes. The ERA5 depth has a significant increasing trend with $p \ll 0.001$, $r^2 = 0.3$, and a slope of 230 m/decade. The CanESM5 amip dataset shows a significant increasing trend, with p = 0.007, $r^2 = 0.2$, and a slope of -71 m/decade. The rest of the datasets do not show any significant depth trend. However, it appears that there may be different trends at select periods; for example, the ERA5 depth appears to suddenly increase after 1992 then decrease for around 12 years, while the depths from the CanESM5 amip data appear to be decreasing over the last decade. On the other hand, the depths from the CanESM5 historical data and the IPSL-CM6A-LR amip data appear to be decreasing near the beginning half of the period. Further analysis will need to be done to determine significant trending periods.

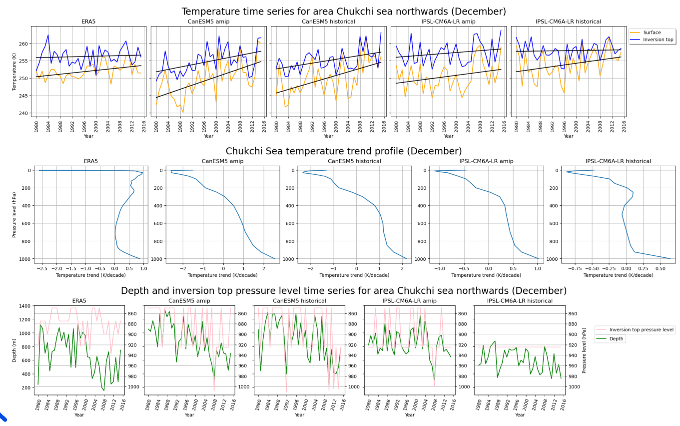

Figure 12: Inversion top and surface time series, trend profile, and depth and inversion top pressure level time series around the Chukchi Sea in December)

The shape of the trend profiles for each dataset generally agree for the lower troposphere, with the temeprature trend decreasing from 1000 hPa to 800 hPa. The CanESM5 datasets show more than double the trend magnitudes than the ERA5 dataset in the region between these pressure levels. The IPSL-CM6A-LR historical dataset agrees well with the ERA5 dataset between these levels, while the IPSL-CM6A-LR amip dataset starts at around the same magnitude as the ERA5 near the surface, then decreases more slowly with elevation.

The ERA5 surface temperature increases significantly at a rate of 0.9 K/decade, with p = 0.006 and $r^2 = 0.2$. Both the top and surface time series for both CanESM5 datasets also have significant trends; for the amip experiment, the surface temperature increases with a slope of 3.0 K/decade, with $p \ll 0.001$ and $r^2 = 0.4$, while the inversion top temperature increases at a rate of 1.7 K/decade, with p = 0.003 and $r^2 = 0.2$. For the historical experiment, the surface temperature increases with a slope of 2.5 K/decade, with $p < 0.001$ and $r^2 = 0.3$, while the inversion top temperature increases at a rate of 1.4 K/decade, with p = 0.004 and $r^2 = 0.2$. The IPSL-CM6A-LR datasets do not result in significant inversion top or surface temperature trends.

All the datasets also result in significant decreasing depth trends over the Chukchi Sea area average, other than IPSL-CM6A-LR amip:

Dataset	**Slope (m/year)**	r^2	**p**
ERA5	-130	0.2	0.004
CanESM5 amip	-160	0.4	2×10^{-5}
CanESM5 historical	-140	0.2	0.003
IPSL-CM6A-LR amip	-37	0.04	0.2
IPSL-CM6A-LR historical	-58	0.1	0.02

Table 1: Chukchi sea northward area average depth trend

References

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Appendix A. Inversion intensity using NaNs

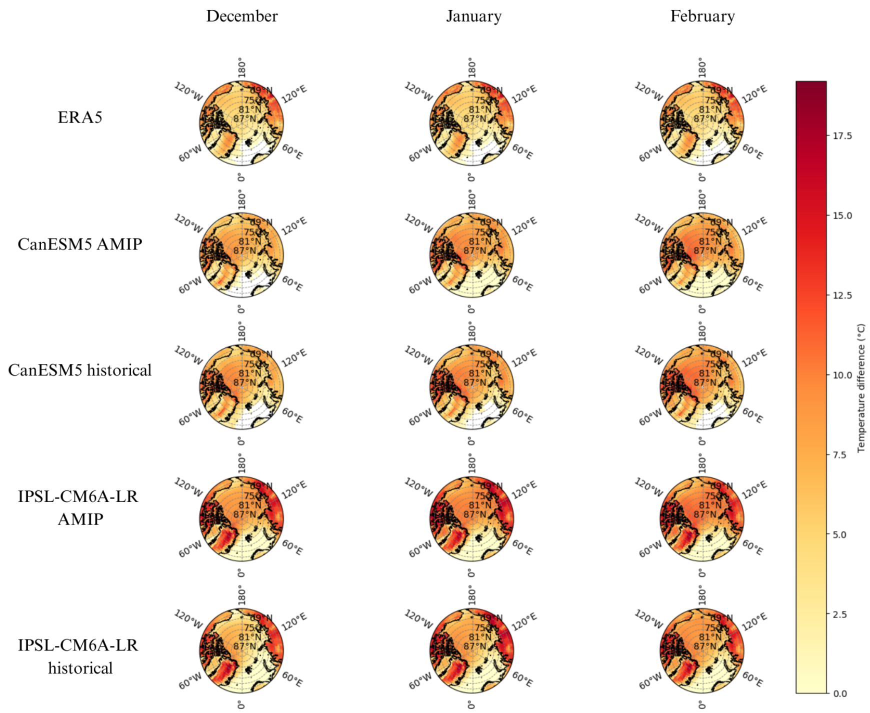

Figure 13: December, January, and February inversion intensity climatologies calculated from each dataset using NaN values for points with no inversion

Appendix B. ERA5 climatologies to 2024

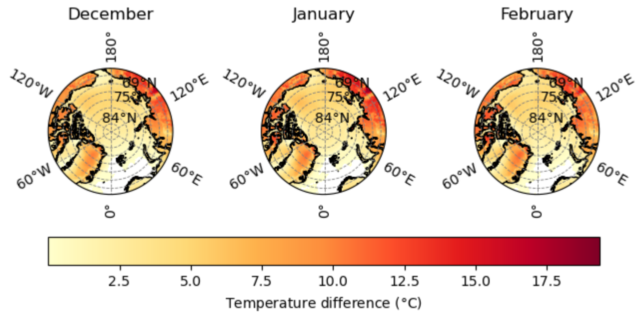

Figure 14: ERA5 1979-2024 average inversion intensities

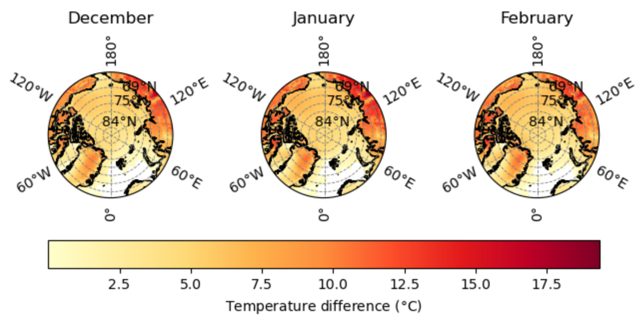

Figure 15: ERA5 1979-2024 average inversion intensities using NaN values for points with no inversion

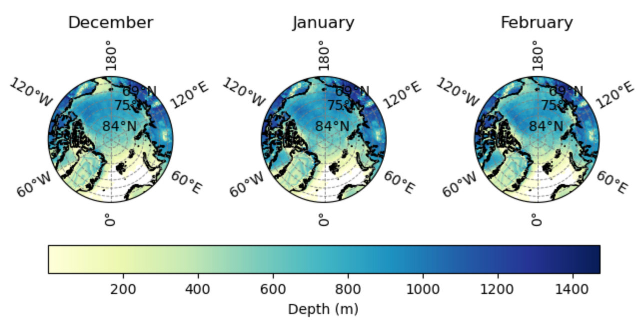

Figure 16: ERA5 1979-2024 average inversion depth